

HDF5_BLS Structure Guide

A Reference for AI Agents

BioBrillouin Project

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1 Introduction

This document is intended for AI agents and automated tools that need to write, parse, or structure Brillouin Light Scattering (BLS) data into HDF5 files compliant with the HDF5_BLS project format. By following this structure and nomenclature, data remains standardized, easily shareable, and fully compatible with the HDF5_BLS suite (Wrapper, Treat, Analyse, and GUI).

2 Core File Hierarchy

An HDF5_BLS-compliant .h5 file is structured hierarchically using groups and datasets. The typical hierarchy is as follows:

```
file.h5
  Brillouin (Root Group)
    Measure 1 (Group)
      Raw_data (Dataset)
      PSD (Dataset)
      Frequency (Dataset)
      Abscissa_0 (Dataset)
      Treatment\_Group\_1 (Group)
        Shift (Dataset)
        Shift\_err (Dataset)
        Linewidth (Dataset)
        Linewidth\_err (Dataset)
        Amplitude (Dataset)
        Amplitude\_err (Dataset)
        Process\_Algorithm.json (Attribute/Text)
    Measure 2 (Group)
    ...
```

2.1 Root Group

By default, the root group is named **Brillouin**. It acts as the container for all subsequent measurements. It **MUST** contain the following attributes:

- **Brillouin_type**: "Root"
- **HDF5_BLS_version**: e.g., "0.1" (Reflects the file structure specification version).

2.2 Dataset Nomenclature

AI agents should give comprehensible names for the datasets that are being stored. The norm does not enforce a fixed nomenclature for groups or arrays, except the top-level group called "Brillouin".

3 The Brillouin_type Attribute

Every group and dataset in an HDF5_BLS-compliant file MUST possess a **Brillouin_type** attribute. This attribute is a string that allows the library and parsing software to recognize the logical category of the element.

3.1 Allowed Group Types

A group's **Brillouin_type** attribute must be one of the following:

- **Root**: Used for organisation groups, that contain other groups.
- **Measure**: Standard measurement group containing experiment raw/PSD data and parameters.
- **Calibration_spectrum**: Group containing calibration spectra data.
- **Impulse_response**: Group containing instrumental impulse response data.
- **Treatment**: Nested subgroup designed specifically to hold data analysis, fitting results, and algorithms.

3.2 Allowed Dataset Types

A dataset's **Brillouin_type** attribute must match one of the following categories:

- **Raw_data**: Raw signal matrices. Must reside within a group of type **Measure**, **Calibration_spectrum**, or **Impulse_response**.
- **PSD**: Power Spectral Density data. Must reside within a group of type **Measure**, **Calibration_spectrum**, or **Impulse_response**.
- **Frequency**: Frequency coordinates corresponding to the PSD.
- **Abcissa_X.Y**: Coordinates for physical dimensions of the measurement (e.g., spatial coordinates). Formatted as **Abcissa_[Dim.start]_[Dim.end]** (e.g., **Abcissa_0_1**).
- **Shift**, **Shift_err**: Brillouin frequency shift fit results and errors. Must reside within a group of type **Treatment**.
- **Linewidth**, **Linewidth_err**: Linewidth (FWHM) fit results and errors. Must reside within a group of type **Treatment**.
- **Amplitude**, **Amplitude_err**: Peak amplitude fit results and errors. Must reside within a group of type **Treatment**.
- **BLT**, **BLT_err**: Brillouin Loss Tangent results and errors. Must reside within a group of type **Treatment**.
- **Other**: General user-defined data arrays that do not map to the categories above.

4 Normalized Attributes

Metadata management is central to the HDF5_BLS philosophy. Attributes must be attached to the appropriate groups (usually the measure group). The **NormalizedAttributes** class defines a standard set of parameters.

AI agents MUST format attribute names as **GROUP.Name** (e.g., **SPECTROMETER.Type**) or store them in a way where the logical group is maintained if handled by the **HDF5_BLS.Wrapper**.

Below is the dictionary of strictly defined attributes:

4.1 MEASURE Attributes

Attributes describing the specific sample and conditions of the measure.

- `MEASURE.Sample`: e.g., "Water"
- `MEASURE.Date_of_measure`: ISO 8601 format (e.g., "2024-11-18T14:17:55.294821")
- `MEASURE.Exposure_(s)`: Exposure time in seconds (e.g., 0.1)

4.2 SPECTROMETER Attributes

Attributes describing the instrumental setup. AI agents should select recommended values when they fit the context.

Attribute Name	Description / Examples
<code>SPECTROMETER.Type</code>	Setup type. <i>Rec</i> : "1 stage VIPA", "2 stage VIPA", "Angle-resolved VIPA", "TFP", "TR-TFP", "TR-1VIPA"
<code>SPECTROMETER.Model</code>	Manufacturer-Model (e.g., "JRS-TFP2")
<code>SPECTROMETER.Wavelength_(nm)</code>	Wavelength in nm (e.g., 532)
<code>SPECTROMETER.Confocal_Pinhole_Diameter_(AU)</code>	Size in Airy Units (e.g., 1)
<code>SPECTROMETER.Detection_Lens_NA</code>	Numerical aperture (e.g., 0.45)
<code>SPECTROMETER.Detector_Model</code>	Manufacturer-Model (e.g., "Hamamatsu-Orca C11440")
<code>SPECTROMETER.Detector_Type</code>	e.g., "CMOS"
<code>SPECTROMETER.Filtering_Type</code>	e.g., "Gas cell"
<code>SPECTROMETER.Filtering_Module</code>	Manufacturer-Model (e.g., "Thorlabs-GC19100-I")
<code>SPECTROMETER.Illumination_Lens_NA</code>	Taking into account beam size before lens/objective (e.g., 0.2)
<code>SPECTROMETER.Illumination_Power_(mW)</code>	Power measured at the sample in mW (e.g., 15)
<code>SPECTROMETER.Illumination_Type</code>	<i>Rec</i> : "CW Laser", "Pulsed Laser"
<code>SPECTROMETER.Laser_Model</code>	Manufacturer-Model (e.g., "Cobolt-Samba")
<code>SPECTROMETER.Laser_Drift_(MHz/h)</code>	Drift measured independently of spectrometer in MHz/h
<code>SPECTROMETER.Phonons_Measured</code>	<i>Rec</i> : "Longitudinal", "Transverse", "Longitudinal & Transverse"
<code>SPECTROMETER.Polarization_probed-analyzed</code>	<i>Rec</i> : "Vertical-Horizontal", "Horizontal-Vertical", "Vertical-Vertical", "Horizontal-Horizontal"
<code>SPECTROMETER.Scan_Amplitude_(GHz)</code>	Only for scanning spectrometers (e.g., 20)
<code>SPECTROMETER.Scanning_Strategy</code>	<i>Rec</i> : "Raster scan", "Line scan", "Point scan"
<code>SPECTROMETER.Scattering_Angle_(deg)</code>	Angle between illumination and detection in degrees (e.g., 180)
<code>SPECTROMETER.Spectral_Resolution_(MHz)</code>	Largest measured value (e.g., 150)
<code>SPECTROMETER.VIPA_FSR_(GHz)</code>	Free Spectral Range of the VIPA (e.g., 30)
<code>SPECTROMETER.x-Mechanical_Resolution_(um)</code>	Estimated mechanical resolution (e.g., 0.5)
<code>SPECTROMETER.x-Optical_Resolution_(um)</code>	Measured optical resolution (e.g., 5)
<code>SPECTROMETER.y-Mechanical_Resolution_(um)</code>	Estimated mechanical resolution (e.g., 0.5)
<code>SPECTROMETER.y-Optical_Resolution_(um)</code>	Measured optical resolution (e.g., 5)
<code>SPECTROMETER.z-Mechanical_Resolution_(um)</code>	Estimated mechanical resolution (e.g., 0.5)
<code>SPECTROMETER.z-Optical_Resolution_(um)</code>	Measured optical resolution (e.g., 50)

4.3 FILEPROP Attributes

Attributes defining file-level properties.

- `FILEPROP.BLS_HDF5_Version`: Version of the spreadsheet nomenclature (e.g., "0.1")

- `FILEPROP.Filepath`: Global absolute or relative name/path of the `.h5` file.
- `FILEPROP.Name`: Human-readable identifier for the file (e.g., "Water spectrum").

5 Implementation Guidelines for AI Agents

When an AI agent is tasked with adding data or translating a workflow into an `HDF5_BLS` structure, it should strictly:

1. **Use the Wrapper methods:** Prefer `add_raw_data()`, `add_PSD()`, `add_frequency()`, `add_abscissa()`, and `add_treated_data()` instead of direct `H5Py` calls. These methods automatically enforce the correct paths and basic attributes.
2. **Apply NormalizedAttributes:** Use `wrp.add_attributes(attributes={...}, parent_group="...")` mapped to the dictionaries expected by the codebase to inject metadata consistently. Never invent new attribute names if an existing normalized attribute applies.
3. **Group treated data logically:** All resulting arrays of a curve fitting or analysis algorithm (shift, linewidth, errors, etc.) must go into an explicit Treatment group under the measured spectra.